



The **Advanced** Materials **Marketplace**

Advanced Materials, Advancing Humanity



Problem

Innovation in Advanced Materials is outpacing adoption due to
“ Knowledge Gaps ”



The cost of inaction

Loss of **leadership** (innovate or perish)
\$164 billion^a lost opportunity (2020-2025)
\$1 trillion operational costs globally due to legacy material & component failure^b



The Buyer's Journey

Does it **exist**?
Does it **solve** my problem?
Where can I **find** it?
How can I **trust** it?
How do I **use** it?



The Valley of Death

15% of global patents don't become products due to a lack of piloting infrastructure^c. Scaling up requires small steps.

a) <https://isg-one.com/articles/smart-manufacturing-accelerates-adoption-post-pandemic-2020-2025>

b) <https://dsindata.co.uk/manufacturing-downtime-costs-and-forecasting/>

c) https://research-and-innovation.ec.europa.eu/document/download/9c492ad5-2c39-4d7c-a6cc-036828ea532a_en?filename=uptake-deployment-advanced-materials.pdf



Solution

NorthStar Platform



AI Assisted Marketplace

A “Problem-First” marketplace that understands the **buyer’s problem** uses a **NorthStar Materials LLM** to create actionable recommendations, faster decisions and scalable adoption, therefore closing the Knowledge Gap.



Our Solution

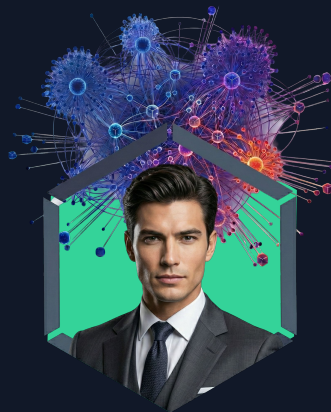
Introducing the **NorthStar Platform**:



**A web platform
where sellers can
upload their products
to buyers**



**Smart recommendations
for which advanced
materials solve problem**



**Agentic AI employees
that actively finds
clients and
ecosystem partners**





Who we are

Dr. Jason Deacon
CEO



7 years of advanced product manufacturing leadership, built and managed 10+ advanced material products, generated \$10M in revenue and lead scientific initiatives of 30+ team members.

Ph.D. Materials U. of Cambridge

Principal Investigator @ SAIT

Lead Scientist @ GLC

Kejvi Peti
CTO



7-year software engineer who's shipped 12 production systems, including a CES 2024 on-device AI agent for smart lighting and a Fortune 500 AI platform for Anheuser-Busch used by millions globally.

MSc in AI U. of Leeds

Anheuser-Busch

Nanoleaf

Rando Kaloti
CGO



8+ years of experience spanning consulting, complex implementations, and leading go-to-market strategy for new products in emerging markets.

Bachelor's of Business, TMU

Sr. Implementation Mgr. @ Dentsu

Consultant @ Deloitte

Kejsi Egro
Dev.





Our Edge



Proprietary Technology

Cutting edge & patentable methodologies to create Knowledge Graphs reducing hallucinations and increasing accuracy.



NorthStar LLM

A tuned LLM with a materials knowledge focus that is trained on advanced materials data.



Team Knowledge

A cumulative 20+ years in product management, 10+ years in advance materials alongside a deep network of advanced material suppliers and users.



Go-to Market Strategy

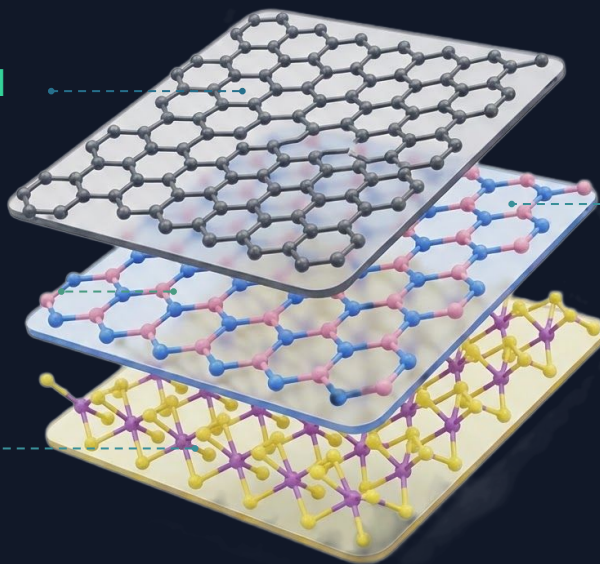
Revenue Model: *Our subscription-based (\$3000/month for AI Employees) & percent-based deal size pricing model (8%-15%) has been received well by our design partners.*

B2B Ecosystem Partnership Model

Institutions like Fraunhofer, The Graphene Flagship, the NRC groups and others produce and support our ICPs. Warm connections there will spread significantly.

Events & Technical Information

Educate others at events and on social media platforms how products featured on our website can be utilized in their specific industrial niche.



Vertical Markets

Advanced materials is vast, we have a monthly outreach program based on what is in high demand and profitable. E.G. Month 1 is Battery Materials, Month 2 is Photovoltaics.

Regions Enabled: Canada, US, Europe



Market



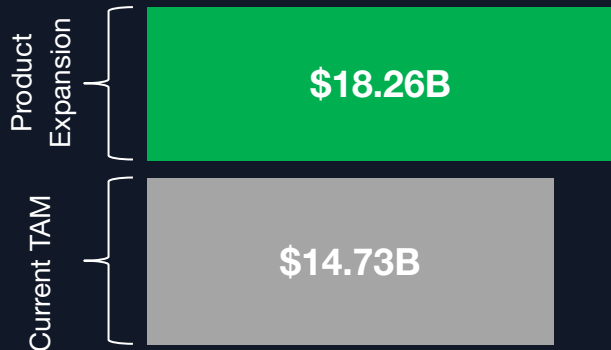
*The global Advanced Materials market is experiencing significant growth and is valued at \$91.27B, while expended to grow at a CAGR of 8.1% until 2030^a. There is a **market trend to using agentic AI for B2B sales, approximate 20% of the industry will use agents by the end of 2026^c.***



ICP: SME clients who produce an advanced material
OR have a product made from advanced materials.



SME: AE, R&D Engineer, Procurement Manager,
Business Developer, Sustainability Officer



TNMI is addressing a \$14.73B market in Canada, US, and Europe that is growing & driven by:



Buyer/Seller Pool: ~7,000 sellers & ~70,000 Buyers
Deal size

~5-25k for initial and
~2 Million annual for integrated product.



Benchmark: Sigma Aldrich makes ~ \$10 billion annually with their marketplace^b. Our goal is 0.1% of that after 2 years and 1-10% after 10 years.

Drivers:



- Technical expertise/skilled labour shortage
- Difficulty sourcing domestic materials
- Market immaturity; opaque pricing + demand
- Global push for technology commercialization

a) <https://www.thebusinessresearchcompany.com/report/advanced-materials-global-market-report>
b) <https://www.merckgroup.com/en/news/q1-2025-15-05-2025.html#:~:text=fully%20consolidated%20entities-,Life%20Science>
c) <https://investor.forrester.com/news-releases/news-release-details/forresters-2026-b2b-marketing-sales-and-product-predictions-b2b/>



Competitors

There are no true competitors that provide a marketplace functionality where buyers & sellers of advanced materials can interact and close deals from a problem-first perspective, because AI has enabled this.

Capabilities		 SIGMA-ALDRICH	MCX
Marketplace	✓	✓	✓
Paid Problem Assistance	✓	✗	✓
Agentic AI Employees	✓	✗	✗
Toll Manufacturers	✓	✗	✗
Ecosystem Creation	✓	✗	✗
AI Problem Assistance	✓	✗	✗



Traction

Client Adoption

Two Letters of Intent and Development Partners secured with COROL (polyol) and Loless (seaweed composite).

User Waitlist

Current waitlist of 20+ users who are excited to be onboarded and utilize our platform.



Venture Lab Collaboration

We have access to Venture Lab's resources and their network in the Critical Industrial Technologies.

Google Startup Program

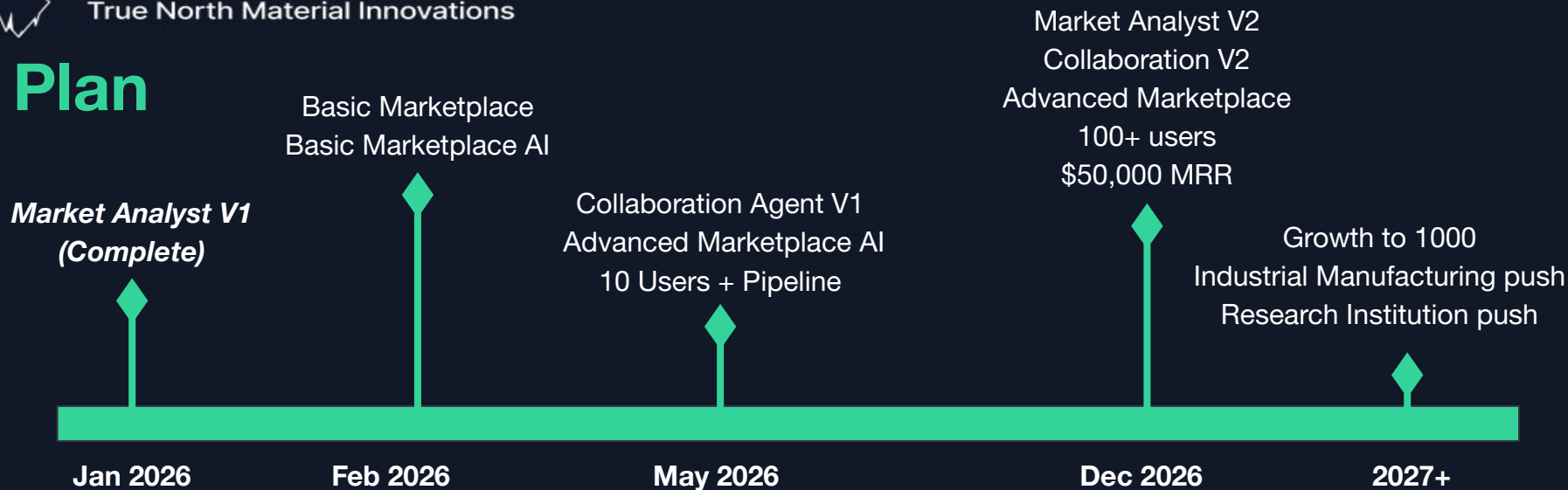
Dedicated support and resources to build.

Product Update

Functional MVP marketplace and AI Employee. Bringing users onto the platform. We need to scale & retain.



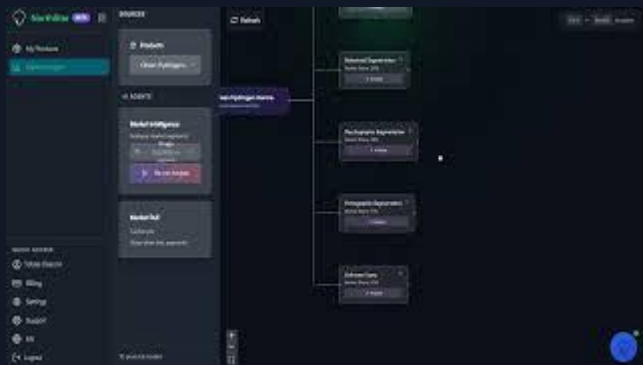
Plan



Category	Allocation (%)	Amount (\$)	Primary Focus
Talent	55%	\$2,200,000	AI/Material Science elite team.
Overhead	20%	\$800,000	Legal, IP, Rent, and Compliance.
Tech/Assets	15%	\$600,000	LLM compute, ASTM testing, Video.
Growth	10%	\$400,000	Beachhead GTM and Partnerships.



Demo for AI Agent



Webinar

Value Chains





True North Material Innovations

THANK YOU

Advanced Materials, Advancing Humanity



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<https://www.linkedin.com/in/kejvipeti/>



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